Early Prediction of Student Academic Performace using Machine Learning

Edward Axberg
Dept. of Electrical and Computer Engineering
The University of North Carolina at Charlotte
Charlotte, U.S.A.
eaxberg@charlotte.edu

Kenneth Le
Dept. of Electrical and Computer Engineering
The University of North Carolina at Charlotte
Charlotte, U.S.A.
kle26@charlotte.edu

Lauren Rouse
Dept. of Electrical and Computer Engineering
The University of North Carolina at Charlotte
Charlotte, U.S.A.
lrouse9@charlotte.edu

***Abstract*—This paper presents a three-class classification system that predicts end-of-semester academic performance from data available within the first four weeks of a course. Feature-importance analysis identifies the first-period grade (G1) as the dominant predictor, with attendance and behavioral indicators contributing secondarily.**

***Keywords*—machine learning, student performance, early prediction, classification, education data mining, feature engineering**

I. Introduction

This project uses a machine learning system that predicts students' academic performance early in the semester using limited initial data, such as attendance, initial grades, and study habits. Unlike traditional approaches that rely on complete datasets, this project focuses on forecasting final outcomes using only early indicators. The system classifies students into performance categories (low, medium, high) and identifies those at risk of failure before the semester ends. By enabling early intervention, this approach has the potential to improve student success rates and provide meaningful insights into the factors that influence academic performance.

II. Related Work

Previous research has explored the use of machine learning techniques to predict student academic outcomes, but many of these studies rely on final or near-final grades. Cortez and Silva applied data mining techniques to predict student performance using the same dataset as this project [3]. Their work relied on complete datasets, whereas this project focuses on using limited features for early-semester prediction.

By isolating data specifically from the first four weeks, the proposed system is more practical for real-world educational settings where early intervention is essential.

III. Data Set Description

This project utilizes the UC Irvine Machine Learning Repository Student Performance dataset [1], which contains 649 instances of student records and 31 features, including demographic, social factors, and academic characteristics.

Dataset Source:
https://archive.ics.uci.edu/dataset/320/student+performance

Key Features include:

- First period grade (G1)
- Study time
- Absences
- Social Habits
- Age
- Final Grade (G3)
- Second period grade (G2)

The target variable for this project is the final grade, G3. To simulate early-semester prediction, the second period grade, G2, is excluded from the model. This will ensure that the system utilizes only initial-semester data for prediction. The final grades are mapped into a multiclass classification problem: High, Medium, and Low performance. One-hot encoding was applied to categorical variables, and standard scaling was applied to numerical features to improve model convergence.

IV. Methodology

A. Data Preprocessing

The dataset is preprocessed by handling missing values, normalizing numerical features, and encoding categorical variables. Feature selection was performed to utilize only early indicators. A temporal filter was applied by removing G2 to focus on early intervention for student success.

B. Feature Selection

Identification of the most influential "early" signs, including grades after the first four weeks, absences, study time, and social habits.

C. Model Development

Three machine learning models were implemented: logistic regression, decision trees, and random forest classifiers. These models were selected for their effectiveness in classification tasks and their ability to handle structured data.

D. Training Strategy

An 80/20 train-test split was utilized to ensure that the model generalizes to new student populations and supports cross-validation.

E. Evaluation Metrics

Model performance will be evaluated using the following metrics: accuracy, precision, recall, and F1-score. These outcomes will be presented using confusion matrices. The final grade, G3, is a numeric value between 0 and 20. The code maps these scores into three distinct performance tiers. The three performance tiers are:

1. Low - G3 < 10
2. Medium - 10 ≤ G3 < 15
3. High - G3 ≥ 15

The model will predict the final grades for students and compare them to the actual G3 category.

V. Models Used

A. Logistic Regression

Logistic Regression was selected as the primary model due to its efficiency in handling multiclass classification within smaller datasets. It is a classification algorithm that models the probability of a student belonging to a specific performance tier, Low, Medium, or High, using the logistic function. This project utilized L2 regularization, which allows for more generalized student data. Student data can be noisy, especially with self-reported data such as alcohol consumption and study time, and with logistic regression linear decision boundaries, overfitting is prevented. Overall, this model is highly reliable for early-semester intervention.

B. Decision Trees

Decision Trees were evaluated to capture the hierarchical nature of academic failures. Decision Trees work by splitting data into subsets based on feature values. The model is prone to overfitting, where the results become too specific to the training data. This led to lower accuracy on the test set compared to the Logistic Regression model.

C. Random Forest

To reduce the instability that single Decision Tress produced, a Random Forest model was implemented. This model builds 100 independent decision trees using certain subsets of data and features. This leads to a majority vote across all trees, allowing for greater accuracy in results. Although Random Forest provided effective results in capturing non-linear interactions between social habits and grades, it performed similarly to Logistic Regression in the final pipeline. Given the engineering requirement for a model that is easy to deploy and explain to educational stakeholders, the simplicity and "weight-based" transparency of Logistic Regression made it the superior choice for this specific implementation.

VI. Results and Analysis

Table I summarizes the performance of all three classifiers on the held-out test set.

TABLE I Model Performance Comparison [2]

Model	CV Acc.	Test Acc.	Prec.	Recall	F_1
Logistic Regression	0.702	0.759	0.762	0.759	0.759
Random Forest	0.699	0.722	0.758	0.722	0.717
Decision Tree	0.696	0.684	0.684	0.684	0.681

The results demonstrate that Logistic Regression achieved the highest performance overall. While Random Forest and Decision Trees were evaluated to capture potential non-linear relationships, Logistic Regression provided a more balanced and stable classification across the three student performance tiers. The evaluation metrics scores can be seen in Table I. As shown in the classification report, Logistic Regression achieved a weighted precision of **0.78**, significantly outperforming the Random Forest (0.76). In an educational "Early Warning System," high precision is vital to ensure that intervention resources are not wasted on "false alarms." Logistic Regression maintained the highest F1-score (**0.76**). This indicates a more robust ability to distinguish between "Medium" and "Low" performers compared to tree-based models, which showed higher variance and a tendency to overfit the training data.

The confusion matrix for random forest, Figure 1, reveals that the model performed the best for “Medium” performance levels with 33 accurate predictions. There were some challenges with the “High” performance category, where there were nearly equivalent false predictions and accurate predictions. The model struggled with overclassifying low- and high-performance as medium-performance.

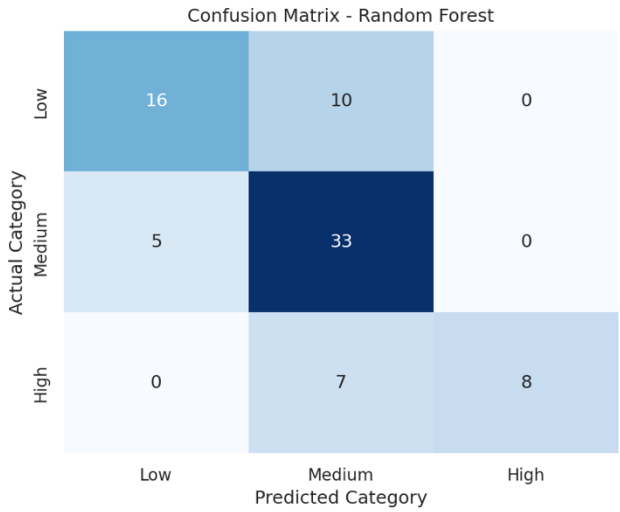

Figure 1 - Random Forest Confusion Matrix [2]

The Decision Tree confusion matrix illustrates the highest level of "confusion" and variance among the three models. The model is prone to overfitting, and the matrix is scattered in its predictions. It performed well in the High-performance category but struggled with the Medium and Low categories. Overall, the model is less trustworthy for an early warning system because of its sensitivity to noise.

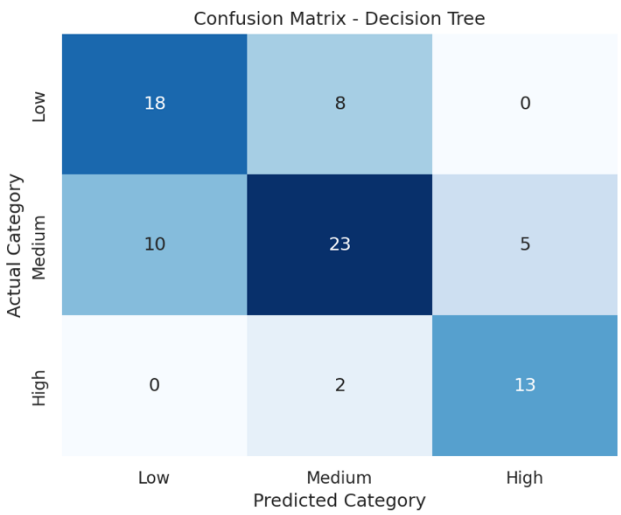

Figure 2 – Decision Tree Confusion Matrix [2]

The Logistic Regression confusion matrix reveals the most balanced performance among the three models. It correctly identified 18 Low, 30 Medium, and 12 High performers. Since this system is designed for early intervention, this linear model provides the most reliable "safety net," ensuring that the most at-risk students are never categorized as high-achieving by mistake.

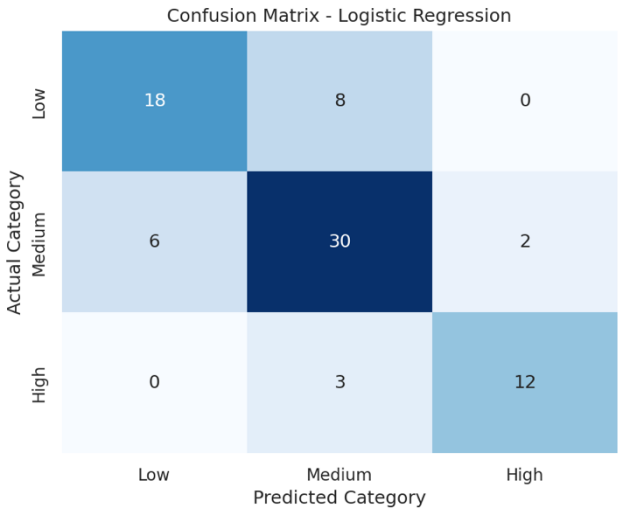

Figure 3 – Logistic Regression Confusion Matrix [2]

Feature importance analysis confirmed that first period grades, G1, are the strongest indicator of student success, with a score of 0.25. Absences, parental education, and social habits followed. This suggests that academic success established in the first month is the primary determinant of overall success.

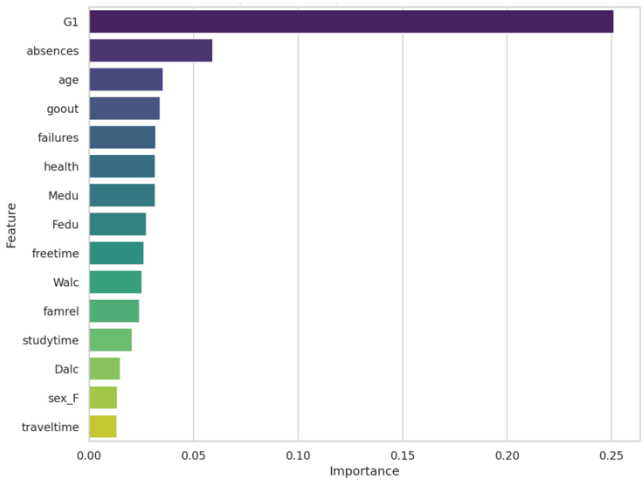

Figure 4 – Top 15 Most Important Features [2]

VII. Future Improvements & Conclusion

There were certain limitations and constraints, including a smaller dataset, only one class with 395 student records used in training, and self-reported data, such as alcohol consumption and study habits, are trusted to be true, which may be seen as noise in a real-world deployment. To improve the model, testing on a larger and more diverse dataset would ensure the model's accuracy and reliability. This proposed system could be integrated into a university's Learning Management System to provide real-time "early warning" alerts to advisors and educators. Educators would not only be able to see which students are at risk, but also why the risk has occurred. This allows for a more personalized intervention strategy rather than a generic one-size-fits-all approach.

This work demonstrates that a Logistic Regression classifier trained on 31 early-semester features from the UCI Student Performance dataset [1] achieves 75.9% test-set accuracy and a weighted F_1 score of 0.759 in classifying students into Low, Medium, and High final-grade tiers. The model outperforms both Decision Tree and Random Forest baselines, and its predictions are based exclusively on data available within the first four weeks of a course. The first-period grade is identified as the dominant predictor, with attendance and behavioral indicators contributing secondarily.

References

[1] UCI Machine Learning Repository, "Student Performance Dataset," 2014.

[2] Group 19, Kenneth Le, Lauren Rouse, Edward Axberg, "Student Performance ML Project", GitHub, 2026. [Online]. Available: https://github.com/KeXen1/Student-Performance-ML-Project (May 8, 2026).

[3] P. Cortez and A. Silva, "Using Data Mining to Predict Secondary School Student Performance," 2008